

To identify the findings by
looking in to your eyes
Case 3 summary

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Gaze picture



Positive findings



- Pupil right 4.5 mm left 3mm
- Anisocoria greater in bright light
- Isolated right third nerve palsy with mild pupil involvement

◉ Work up

- Vasculopathic risk factor : DM, DLP

◉ Ddx

- Ischemic
- Compressive : pupil involvement

Lesions of the third nerve nucleus

- ◉ Rare
- ◉ Unilateral third nerve palsy
- ◉ Contralateral superior rectus weakness
- ◉ Bilateral ptosis

Fascicular lesion

- ◉ Distal to nucleus but within the brainstem
 - ◉ Impact many structure
 - ◉ Multiple deficits
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- ◉ Ex. Weber syndrome : cerebral peduncle
Benedikt syndrome : red nucleus

Cavernous sinus

- More susceptible to compression against the interclinoid ligaments above and the petroclinoid ligament below
- Involve other structure in cavernous sinus

Orbital portion

- ◉ Before entering of the SOF, split in 2 divisions
 - Superior division : SR, LP
 - Inferior division : IR, MR, IO pupil
- ◉ Proptosis, lid swelling, conjunctival injection, and chemosis

Ischemic isolated third nerve palsy

- ◉ 10 – 20% affect pupil
- ◉ Complete EOM palsy with normal pupil size with vasculopathic risk factor
 - May observe without neuroimaging
- ◉ Partial third nerve palsy without pupillary involvement
 - Must re-evaluate within 1 week for pupillary abnormality

Cause of isolated ocular motor nerve palsy

- 109 pts aged ≥ 50 years with acute isolated ocular motor nerve palsy
 - 22 : CN III palsy
 - 25 : CN IV palsy
 - 62 : CN VI palsy
- 18 patients cause other than presumed microvascular ischemia

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- Presence ≥ 1 vasculopathic risk factors
 - Significantly asso. with a presumed microvascular cause ($P = 0.003$)
 - Other causes group
 - 61% (11/18) Vasculopathic risk factors
 - Vasculopathic risk factors group
 - 10% (8/80) : other causes
 - Midbrain infarction, neoplasms, inflammation, pituitary apoplexy, and giant cell arteritis (GCA)

Aspirin use and the prevention of acute ischemic cranial nerve palsy

- Retrospective case-control study
- 100 pts with ischemic cranial nerve palsies
- Association with diabetes, hypertension, or both (palsy cases)
- 163 age-matched and sex-matched patients

Aspirin use and the prevention of acute ischemic cranial nerve palsy

- Aspirin not associated with a reduced **rate of ischemic** third, fourth, sixth, and seventh nerve palsies among patients with diabetes mellitus and hypertension.
- Aspirin appears to be ineffective in preventing ischemic third, fourth, sixth, and seventh cranial nerve palsies

Isolated CN 3 palsy

Age<50

Brain
MRI/MRA/CTA

LP+/-

Cerebral
angiogram+/-

Age>50

Pupil
sparing

Complete
palsy

FBG,ESR,
CRP

MG

Closed
observe

Partial
palsy

MRI/MRA/
CTA

FBG,ESR,
CRP

Catheter
angiography+/-

Pupil
involved

MRI/MRA/
CTA

THANK YOU